

Mark schemes

Q1.

(a) (i) $(x - 3)^2 + (y + 5)^2$

One term correct

B1

LHS correct with + and squares

B1

$$= 25 - 9 + 9 = 25 \quad (= 5^2)$$

Condone RHS = 25

B1

3

(b) (i) $C(3, -5)$

B1ft

(ii) Radius = 5

Correct or ft their RHS provided > 0

B1ft

2

(c) (i) $(7 - 3)^2 + (-2 + 5)^2 = 16 + 9 = 25$

Or sub'n of (7, -2) in original equation

$\Rightarrow D$ lies on circle

Must see statement

$$7^2 + (-2)^2 - 42 - 20 + 9 = 0$$

Or sub $x = 7$ into eqn & showing $y = -2$ etc

B1

1

(ii) Attempt at gradient of CD as normal grad $CD =$

$$\frac{-2 - (-5)}{7 - 3} = \frac{3}{4}$$

withhold if subsequently uses $m_1 m_2 = -1$

$$\frac{\Delta y}{\Delta x}$$

(condone one slip) FT their centre C

M1

$$y + 2 = \frac{3}{4}(x - 7) \quad \text{or} \quad y + 5 = \frac{3}{4}(x - 3)$$

Correct equation in any form $y = \frac{3}{4}x - \frac{29}{4}$

A1

$$\Rightarrow 3x - 4y = 29$$

CSO **Integer** coefficients
 Condone $4y - 3x + 29 = 0$ etc

A1

3

- (d) (i) $y = kx$ sub'd into original circle equation
 or using their completed square form and
 multiplying out

M1

$$x^2 + (kx)^2 - 6x + 10kx + 9 = 0$$

$$\Rightarrow (k^2 + 1)x^2 + 2(5k - 3)x + 9 = 0 \quad \text{AG}$$

CSO
 must see at least previous line for A1
 any error such as $kx^2 = \dots = k^2 x^2$ gets A0

A1

2

(ii) $4(5k - 3)^2 - 36(k^2 + 1)$

Discriminant in k (can be seen in quad
 formula)
 Condone one slip

M1

$$= 64k^2 - 120k$$

$$\text{or } 8k^2 - 15k = 0 \quad \text{OE}$$

A1

$$\text{Equal roots: } 4(5k - 3)^2 - 36(k^2 + 1) = 0$$

$b^2 - 4ac = 0$ clearly stated or evident by
 an equation in k with at most 2 slips.

B1

$$8k^2 - 15k = 0$$

Attempt to solve **their** quadratic or linear
 equation if k has been cancelled

m1

$$\Rightarrow k = 0, k = \frac{15}{8}$$

OE but must have $k = 0$

If “= 0” is not seen but correct values of k are found, candidate will lose B1 mark but may earn all other marks

A1

5

- (iii) (Line is a) **tangent** (to the circle)
*Line **touches** circle at one point*

E1

1

[17]

Q2.

(a) (i) $y = -\frac{3}{5}x + \frac{11}{5}$

Or correct expression for gradient using two correct points

Attempt at $y = f(x)$

M1

Or answer $= \frac{3}{5}$ or $-\frac{3}{5}x$ gets M1

But answer of $\frac{3}{5}x$ gets M0

(Gradient of $AB = -\frac{3}{5}$)

Correct answer scores 2 marks.

Condone error in rearranging formula if answer for gradient is correct.

A1

2

(ii) $m_1 m_2 = -1$

Used or stated

M1

Gradient of perpendicular = $\frac{5}{3}$

ft their answer from (a) (i) or correct

A1ft

$$y - 1 = \frac{5}{3}(x - 2)$$

OE

$$5x - 3y = 7; \text{ or } y = \frac{5}{3}x + c, \quad c = -\frac{7}{3} \text{ etc}$$

CSO

A1

3

(b) Eliminating x or y but must use

$$3x + 5y = 11 \quad \& \quad 2x + 3y = 8$$

An equation in x only or y only

M1

$$x = 7$$

A1

$$y = -2$$

A1

Answer only of $(7, -2)$ scores 3 marks

3

[8]

Q3.

(a) (i) $C(5, -12)$

B1

1

(ii) Radius = 13 (or $\sqrt{169}$)

$\pm \sqrt{169}$ or ± 13 as final answer scores B0

B1

1

(b) (i) $(-5)^2 + 12^2$ or $25 + 144$

$$= 169 \Rightarrow \text{circle passes through } O$$

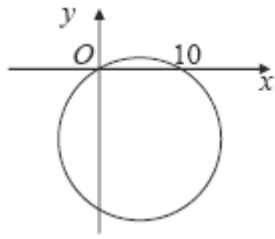
Correct arithmetic plus statement

Eg " O lies on circle", "as required" etc

B1

1

(ii) Sketch



*Freehand circle through origin and cutting
 positive x-axis with centre in 4th quadrant
 Condone value 10 missing or incorrect*

B1

$$25 + (p + 12)^2 = 169$$

Or doubling their y_C -coordinate

M1

$$(p + 12) = \pm 12 \quad p = -24$$

Condone use of y instead of p

*SC B2 for correct value of p stated or
 marked on diagram*

A1

3

(c) (i) $\text{grad } AC = \frac{-12 + 7}{5 + 7}$

correct expression, but ft their C

M1

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$$= -\frac{5}{12}$$

Condone $\frac{5}{-12}$

A1

2

(ii) $\text{grad tangent} = \frac{12}{5}$
 $\frac{-1}{\text{their grad } AC}$

B1ft

$$y + 7 = \frac{12}{5}(x + 7)$$

ft " $\frac{12}{5}$ " must be **tangent** and not AC

M1

$$\Rightarrow 12x - 5y + 49 = 0$$

OE with **integer** coefficients with all terms on one side of the equation

A1

3

[11]

Q4.

(a) $\frac{dx}{dt} = \frac{1}{t^2} \quad \frac{dy}{dt} = 1 - \frac{1}{2t^2}$

B1B1

$$\frac{dy}{dx} = \frac{1 - \frac{1}{2t^2}}{-\frac{1}{t^2}} \left(= \frac{2t^2 - 1}{-2} \right)$$

$\frac{\frac{dy}{dt}}{\frac{dx}{dt}}$
 Their $\frac{dy}{dt}$; condone 1 slip

M1

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CSO; ISW

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A1

Alternative

$$y = \frac{1}{x} + \frac{x}{2}$$

(B1)

$$\frac{dy}{dx} = -\frac{1}{x^2} + \frac{1}{2}$$

(B1)

Substitute $x = \frac{1}{t}$

(M1)

$$\frac{dy}{dx} = -t^2 + \frac{1}{2}$$

CSO

(A1)

4

(b) $t = 1 \quad \frac{dy}{dx} = \frac{1}{2}$

Substitute $t = 1$ in $\frac{f(t)}{g(t)} \neq k$

M1

$$m_T = -\frac{1}{2} \Rightarrow m_x = 2$$

F on $m_T \neq 0$; if in $t \rightarrow$ numerical later

B1F

$$(x, y) = (1, \frac{3}{2})$$

$$Pl \frac{3}{2} = m(x1) + c$$

B1

$$(y - \frac{3}{2}) = 2(x - 1) \text{ or } y = 2x + c, c = -\frac{1}{2}$$

ISW, CSO (a) and (b) all correct

A1

4

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$$y = \frac{1}{1} + \frac{1}{2} \times \frac{1}{t}$$

(c)

*Attempt to use $t = \frac{1}{x}$ to eliminate t
*t, or equivalent**

M1

$$= \frac{1}{x} + \frac{x}{2}$$

A1

$$2xy = 2 + x^2 \Rightarrow x^2 - 2xy + 2 = 0$$

Correct algebra to AG with $k = 2$

allow $k = 2$ stated $k = 2$,

no working or from $\left(1, \frac{1}{3}\right): 0/3$

A1

Alternative

or

$$\left(\frac{1}{t}\right)^2 - 2\left(\frac{1}{t}\right)\left(t + \frac{1}{2t}\right) \quad \left| \quad xy = \frac{1}{t}\left(t + \frac{1}{2t}\right) \right.$$

$$\quad \quad \quad = -2 \quad \quad \quad \quad \quad = 1 + \frac{x^2}{2}$$

Substitute and multiply out

(M1)

Eliminate t

(A1)

$$\Rightarrow x^2 - 2xy + 2 = 0$$

Conclusion, $k = 2$

(A1)

3

[11]

Q5.

- (a) Mid-point of $BC = (3, -2)$

Either coordinate correct

B1

Both cords correct. Accept $x = 3, y = -2$

B1

2

(b) (i) $\frac{\Delta y}{\Delta x} = \frac{3-1}{-2-4}$

$$\pm \frac{2}{6} \quad \text{OE implies M1}$$

M1

$$= -\frac{1}{3}$$

A1

2

- (ii) $y - 3 = \text{"their grad"} (x + 2)$ or
 $y - 1 = \text{"their grad"} (x - 4)$
Or $y = mx + c$ and correct attempt to find c

M1

Hence $x + 3y = 7$

A1

2

- (iii) $y + 5 = \text{"their grad } AB" (x - 2)$
Or " $their x + qy = c$ " and attempt to find c

M1

$$y + 5 = -\frac{1}{3}(x - 2) \text{ or } x + 3y + 13 = 0$$

OE

A1

2

- (c) Grad $BC = 3$ (from $\frac{\Delta y}{\Delta x} = \frac{1+5}{4-2}$ OE)
Or 2 lengths correct:
 $AB = \sqrt{40}; BC = \sqrt{40}; AC = \sqrt{80}$

B1

$m_1 m_2 = -1$ stated **or**

grad $BC = 3$ and grad $AB = -\frac{1}{3}$ **or**

grad $BC \times$ grad $AB (= 3 \times -\frac{1}{3})$

Or attempt at Pythagoras or Cosine Rule

M1

Product of gradients = -1

Hence AB and BC are perpendicular

Completing proof and statement

A1CSO

3

[11]

Q6.

(a) (i) $p(-1) = -1 + 7 - 6$
Finding $p(-1)$

M1

$= 0$ therefore $x + 1$ is a factor
*Shown to = 0 **plus statement***

A1

2

(ii) $p(x) = (x + 1)(x^2 - x - 6)$
Long division/inspection (2 terms correct)

M1

Quadratic factor correct

A1

$p(x) = (x + 1)(x + 2)(x - 3)$
*May earn M1, A1 For correct second factor
 then A1 For $(x + 1)(x + 2)(x - 3)$*

A1

3

(b) (i) $A(-2, 0)$
Condone $x = -2$

B1

1

(ii) $\frac{x^4}{4} - \frac{7x^2}{2} - 6x (+c)$
One term correct

M1

Another term correct

A1

(may have + c or not)
All correct unsimplified

A1

$\left[\frac{81}{4} - \frac{63}{2} - 18 \right] - \left[\frac{1}{4} - \frac{7}{2} + 6 \right]$

$F(3) - F(-1)$ attempted in correct order

m1

$$= -32$$

CSO; OE

A1

5

(iii) Area of shaded region = 32

FT their (b)(ii) but positive value needed

B1ft

1

(iv) $\frac{dy}{dx} = 3x^2 - 7$

One term correct

M1

All correct (no + c etc)

A1

When $x = -1$, gradient = -4

CSO

A1

3

(v) Gradient of normal = $\frac{1}{4}$

B1ft

$y = \text{"their gradient"} (x \pm 1)$

*Must be finding **normal**, not tangent*

M1

$$y = \frac{1}{4}(x + 1)$$

CSO; any correct form eg $4y - x = 1$

A1

3

[18]

Q7.

(a) (i) $\frac{dy}{dx} = 4 \times \frac{1}{2} x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}} = 2x^{-\frac{1}{2}} - \frac{3}{2} x^{\frac{1}{2}}$

A power decreased by 1

M1

A1 For each correct term

A1 A1

3

(ii) At $P(4, 0)$, $\frac{dy}{dx} = \frac{2}{\sqrt{4}} - \frac{3}{2} \times 2$

Attempts $\frac{dy}{dx}$ when $x = 4$

M1

$= 1 - 3 = -2$

AG

A1

2

(iii) Gradient of normal = $-\frac{1}{2}$

Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 0 = m [x - 4]$

m numerical; can be awarded even if

$m = -2$

M1

$y - 0 = \frac{1}{2} (x - 4) \Rightarrow 2y = x - 4$

ACF of the equation

A1

3

(iv) At Q , $x = 0$, $2y = 0 - 4$

PI

M1

$y_e = -2$

Area of triangle $OPQ = 0.5 \times 4 \times |y_e|$

Ft on a linear equation for normal provided y_e is negative and prev A1 is lost

A1F

$= 4$

Ft on c's negative y_e

B1F

3

(v) $2x^{-\frac{1}{2}} - \frac{3}{2}x^{\frac{1}{2}} = 0 \Rightarrow 2x^{-\frac{1}{2}} = \frac{3}{2}x^{\frac{1}{2}}$

Puts c's $\frac{dy}{dx} = 0$ and a 1st step in attempt to solve.

M1

$2 = \frac{3}{2}x \Rightarrow x = \frac{4}{3}$

Valid method to $ax = b$

m1

Condone 1.3 or better

A1

3

(b) (i) $\int \left(4x^{\frac{1}{2}} - x^{\frac{3}{2}} \right) dx = 4 \frac{x^{\frac{3}{2}}}{1.5} - \frac{x^{\frac{5}{2}}}{2.5} \{+c\}$
 $= \frac{8}{3}x^{\frac{3}{2}} - \frac{2}{5}x^{\frac{5}{2}} \{+c\}$

One power correct

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M1

Condone absence of "+ c"

A1,A1

3

(ii) Area under curve = $4 \frac{4^{\frac{3}{2}}}{1.5} - \frac{4^{\frac{5}{2}}}{2.5} - \{0\}$
 $F(4) - \{F(0)\}$

M1

Total area = $F(4)$ + area triangle OPQ

m1

Total area = $\frac{128}{15} + 4 = \frac{188}{15} = 12.5(3...)$

Accept 3 sf if clear

A1

3

[20]

Q8.

(a) $\text{Grad } AC = \frac{15}{3} = 5$
OE

B1

Equation of AC: $y = m(x + 2)$
or $(y - 15) = m(x - 1)$
Or use of $y = mx + c$ with $(-2, 0)$ or
 $(1, 15)$ correctly substituted for x and y

M1

$y = 5x + 10$

OE eg $y - 15 = 5(x - 1)$, $y = 5(x + 2)$

A1

3

(b) (i) $\left[16x - \frac{x^5}{5}\right]$

Raise one power by 1

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M1

One term correct

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A1

All correct

A1

$\left(16 - \frac{1}{5}\right) - \left(-32 + \frac{32}{5}\right)$

F(1) - F(-2) attempted

m1

$= 41\frac{2}{5}$ (or 41.4, $\frac{207}{5}$ etc)

CSO; withhold if + c added

A1

5

(ii) Area $\Delta = \frac{1}{2} \times 3 \times 15$ or $22\frac{1}{2}$ or 22.5

Or $\int_{-2}^1 (5x + 10) dx = 22.5$

B1

Shaded area = "their (b)(i) answer" – correct triangle
 Condone "difference" if $\Delta > \int$

M1

$\Rightarrow$ shaded area = $18\frac{9}{10}$
 CSO; OE (18.9 etc)

A1

3

[11]

Q9.

(a) $(x - 8)^2 + (y - 13)^2$

Exactly this with + and squares

B1

= 13^2

Condone 169

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B1

2

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(b) (i) grad $PC = \frac{12}{5}$

Must simplify $\frac{-12}{-5}$

B1

1

(ii) grad of tangent = $\frac{-1}{\text{grad } PC} = -\frac{5}{12}$

Condone $-\frac{1}{2.4}$ etc

B1ft

tangent has equation $y - 1 = -\frac{5}{12}(x - 3)$
 ft gradient but M0 if using grad PC

M1

Correct – but not in required final form

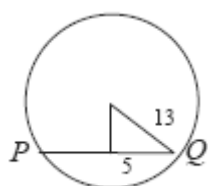
A1

$5x + 12y = 27$ OE
 MUST have integer coefficients

A1

4

(iii)



half chord = 5
 Seen or stated

B1

$d^2 = (\text{their } r)^2 - 5^2$
 (provided $r > 5$)
 Pythagoras used correctly $d^2 = 13^2 - 5^2$

M1

Distance = 12
 CSO

A1

3

[10]

Q10.

(a) $\sqrt{x^3} = x^{\frac{3}{2}}$

OE; accept 'k = 1.5'

B1

1

(b) (i) $\frac{dy}{dx} = 2x - \frac{3}{2}x^{\frac{1}{2}}$

At least one index reduced by 1 and no term of the form $\sqrt{ax^2}$.

M1

For $2x$

B1

For $-1.5x^{0.5}$. Ft on ans (a) non-integer k

A1F

3

(ii) When $x = 4$, $y = 8$

B1

$y'(4) = ;$

Attempt to find $\frac{dy}{dx}$ when $x = 4$

M1

$= 2(4) - 1.5(\sqrt{4}) = 5$

Ft on one earlier error provided noninteger powers in (a) and (b) (i)

A1F

Tangent: $y - 8 = 5(x - 4)$

$y - y(4) = y'(4)[x - 4]$ OE

m1

$y = 5x - 12$

CSO; must be $y = 5x - 12$

A1

5

[9]

Q11.

(a) $\frac{dx}{dt} = 4$ $\frac{dy}{dt} = -\frac{1}{2t^2}$

differentiate. 4; at^{-2} seen

M1

both derivatives correct

A1

$$\frac{dy}{dx} = -\frac{1}{2t^2} \times \frac{1}{4}$$

use chain rule candidates' $\frac{dy}{dt} / \frac{dx}{dt}$

M1

$$t = \frac{1}{2} \quad \frac{dy}{dx} = -\frac{1}{2}$$

CSO

A1

ALTERNATIVE METHOD:

$$y = \frac{2}{x-3} - 1 \quad \text{and differentiate}$$

differentiate expression in y and x

$$\frac{dy}{dx} = \frac{-2}{(x-3)^2}$$

correct

M1

A1

$$x = 5$$

$$\frac{dy}{dx} = \frac{-2}{(5-3)^2}$$

find and therefore use x (and y)

m1

$$\frac{dy}{dx} = -\frac{1}{2}$$

A1

4

(b) gradient of normal = 2

F if gradient $\neq \pm 1$

B1F

$$(x, y) = (5, 0) \quad \frac{y}{x-5} = 2$$

calculate and use (x, y) on normal

M1

F on gradient of normal ACF

A1F

3

$$(c) \quad x - 3 = 4t \quad \text{or} \quad y + 1 = \frac{1}{2t}$$

$$\text{or} \quad t = \frac{x-3}{4} \quad \text{or} \quad \frac{1}{t} = 2(y+1)$$

B1

$$(x - 3)(y + 1) = 2$$

eliminate t ; allow one error

M1

$$y = \frac{\frac{1}{2(x-3)} - 1}{4} \quad \text{ACF}$$

accept

SC allow marks for part (c) if done in part (a)

A1

3

[10]

Q12.

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$$(a) \quad (i) \quad y = -\frac{3}{5}x + \dots; \quad \text{Gradient } AB = -\frac{3}{5}$$

Attempt to find $y =$ or $\Delta y / \Delta x$ or $\frac{3}{5}$ or $3x/5$

M1

Gradient correct – condone slip in $y = \dots$

A1

2

$$(ii) \quad m_1 m_2 = -1$$

Stated or used correctly

M1

Gradient of perpendicular = $\frac{5}{3}$
ft gradient of AB

A1 ✓

$$\Rightarrow y + 2 = \frac{5}{3}(x - 6)$$

CSO Any correct form eg $y = \frac{5}{3}x - 12$,
 $5x - 3y = 36$ etc

A1

3

(b) Eliminating x or y (unsimplified)

Must use $3x + 5y = 8$; $2x + 3y = 3$

M1

$$x = -9$$

A1

$$y = 7$$

$B(-9, 7)$

A1

3

(c) $4^2 + (k + 2)^2 (=25)$ or $16 + d^2 = 25$
Diagram with 3, 4, 5 triangle

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M1

$$k = 1$$

A1

Condone slip in one term (or $k + 2 = 3$)

or $k = -5$

SC1 with no working for spotting one correct value of k . Full marks if both values spotted with no contradictory work

A1

3

[11]

Q13.

(a) (i) $B(0, 5)$

B1

Area AOB $\frac{1}{2} \times 1 \times 5$

Condone slip in number or a minus sign

M1

$$= 2\frac{1}{2}$$

A1

3

(ii) $\frac{3x^6}{6} + \frac{2x^2}{2} + 5x$ or $\frac{x^6}{2} + x^2 + 5x$ (may have + c or not)

Raise one power by 1

M1

One term correct

A1

All correct unsimplified

A1

3

(iii) Area under curve = $\int_{-1}^0 f(x) dx$
Correctly written or $F(0) - F(-1)$ correct

B1

$$[0] - \left[\frac{1}{2} + 1 - 5 \right]$$

*Attempt to sub limit(s) of -1 (and 0)
 Must have integrated*

M1

Area under curve $3\frac{1}{2}$
CSO (no fudging)

A1

$$\text{Area of shaded region} = 3\frac{1}{2} - 2\frac{1}{2} = 1$$

FT their integral and triangle (very generous)

B1 ✓

4

(b) (i) $\frac{dy}{dx} = 15x^4 + 2$

One term correct

M1

All correct (no +c etc)

A1

when $x = -1$, gradient = 17

CSO

A1

3

(ii) $y = \text{"their gradient"} (x + 1)$

Must be finding **tangent** – not normal
any form e.g. $y = 17x + 17$

B1 ✓

1

[14]

Q14.

(a) (i) $y = x + 1 + 4x^{-2} \Rightarrow \frac{dy}{dx} = 1 - 8x^{-3}$

M1

Power $p \rightarrow p - 1$

(A1 if $1 + ax^n$ with $a = -8$ or $n = -3$)

A2,1,0

3

(ii) $1 - 8x^{-3} = 0$

Puts c's $\frac{dy}{dx} = 0$

M1

$x^3 = 8$

Using $x^{-k} = \frac{1}{x^k}$ to reach $x^a = b$, $a > 0$ or correct use of logs.

m1

$$x = 2$$

A1

When $x = 2$, $y = 4$

A1ft

4

(iii) At (1, 6), $\frac{dy}{dx} = 1 - 8 = -7$
Attempt to find $y'(1)$

M1

Gradient of normal = $\frac{1}{7}$
Use of or stating $m \times m' = -1$

M1

Equation of normal is $y - 6 = m[x - 1]$
m numerical

M1

$$y - 6 = \frac{1}{7}(x - 1)$$

$$\left\{ \frac{y-6}{x-1} = \frac{1}{7}; 7y = x + 41 \right\}$$

OE ft on c's answer for (a)(i) provided at least A1 given in (a)(i) and previous 3M marks awarded

Alft

4

(b) (i) $\int x \left(1 + \frac{4}{x^2} \right) dx =$

$$\dots\dots\dots = \frac{x^2}{2} + x - 4x^{-1} \{+ c\}$$

One of three terms correct.

M1

*One of three terms correct.
 For A2 need all three terms as printed or better*

(A1 if 2 of 3 terms correct)

A2,1,0

3

(ii) {Area = } $\int_1^4 x + 1 + \frac{4}{x^2} dx =$
 $\left[\frac{x^2}{2} + x - \frac{4}{x} \right]_1^4 = (8 + 4 - 1) - \left(\frac{1}{2} + 1 - 4 \right)$

*Dealing correctly with limits; F(4)–F(1)
(must have integrated)*

M1

= 13.5

A1

2

[16]

Q15.

(a) (i) $y = (4x^2 + 3x + 2)^{10}$
 $\frac{dy}{dx} = 10(4x^2 + 3x + 2)^9 (8x + 3)$

M1

For f(x)()⁹ where f(x) ≠ k and is linear

A1

2

(ii) $y = x^2 \tan x$
Product rule

M1

$\frac{dy}{dx} = x^2 \sec^2 x + 2x \tan x$

A1

2

(b) (i) $x = 2y^3 + \ln y$

$\frac{dx}{dy} = 6y^2 + \frac{1}{y}$

B1

(ii) At (2,1)

$$\frac{dx}{dy} = 6 + 1 = 7$$

M1

$$\frac{dy}{dx} = \frac{1}{7}$$

May be implied

A1 ✓

$$(y-1) = \frac{1}{7}(x-2)$$

OE

A1

3

[8]

Q16.

(a) (i) $\int (4 - e^{2x}) dx$
 $= 4x - \frac{1}{2} e^{2x} (+c)$

4x

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$$-\frac{1}{2} e^{2x}$$

B1

B1

2

(ii) $\int_0^{\ln 2} \left[4x - \frac{1}{2} e^{2x} \right] dx$
 $= \left[4 \ln 2 - \frac{1}{2} e^{2 \ln 2} \right] - \left[(0) - \frac{1}{2} (e^0) \right]$

Substitute both ln 2 and 0 correctly into an integrated expression

M1

$$= 4 \ln 2 - 2 + \frac{1}{2}$$

Convincing

$$= 4 \ln 2 - \frac{3}{2}$$

AG

A1

2

(b) (i) $x = 0$

$$y = 4 - 1 = \underline{\underline{3}}$$

B1

1

(ii) At B, $y = 0$

$$4 - e^{2x} = 0$$

Or reverse argument

$$e^{2x} = 4$$

$$x = \ln 2$$

AG

M1

A1

2

(c) $\frac{dy}{dx} = -2e^{2x}$

B1

$$x = \ln 2, \text{ Gradient} = -2e^{2 \ln 2} = -8$$

$$x = \ln 2 \text{ into } ke^{2x}$$

M1

$$\text{Gradient normal} = \frac{1}{8} = \frac{1}{2e^{2 \ln 2}}$$

OE

A1

Equation $y = -\frac{1}{8}x - \frac{1}{8}\ln 2$
OE

A1

4

(d) When $x = 0$

$$y = -\frac{1}{8}\ln 2$$

*Attempt to integrate their line and substitute
 $x = 0, \ln 2$*

M1

Area $\Delta = \frac{1}{16}(\ln 2)^2$ condone – ve sign = 0.03

$$\frac{1}{2}(\text{their } y) \times \ln 2$$

A1 ✓

Total area $4\ln 2 - \frac{3}{2} + \frac{1}{16}(\ln 2)^2 = 1.30$
CSO

AWRT

A1

3

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[14]

Q17.

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(a) (i) $\frac{dx}{dt} = 2, \quad \frac{dy}{dt} = -8t$
CAO

B1, B1

2

(ii) $\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{-8t}{2} = -4t$

M1

*Chain rule in correct form
ft on sign coefficient errors (**not** power of t)*

A1F

2

(b) $m_T = -4, m_N = \frac{1}{4}$

ft on $\frac{dy}{dx}$ if $f(t)$

B1F,
B1F

$x = 3 \quad y = -3$

$$\frac{y-3}{x-3} = \frac{1}{4} \Rightarrow \frac{y+3}{x-3} = \frac{1}{4}$$

Use candidate's (x, y) and m_N

M1

Any correct form; ISW; CAO

A1

4

(c) $t = \frac{x-1}{2}$

$$y = 1 - 4 \left(\frac{x-1}{2} \right)^2$$

Substitute for t

Simplification not required but CAO

Or equivalent methods / forms:

$$y = 2x - x^2, \quad t^2 = \frac{1-y}{4},$$

$$\left(\frac{x-1}{2} \right)^2 = \frac{1-y}{4}$$

M1

M1A1

3

[11]

Q18.

(a) $\lg y = \lg a + b \lg x$

M1 for use of one log law

M1A1

2

(b) Use of above result

M1

$$a = 10$$

A1

$$b = \text{gradient}$$

$$OE; PI \text{ by answer } \pm \frac{1}{2}$$

m1

$$\dots = -\frac{1}{2}$$

A1

4

[6]

Q19.

$$(a) \quad (i) \quad \text{Gradient } AB = \frac{-1-5}{6-2} \text{ or } \frac{5--1}{2-6}$$

$$\pm \frac{6}{4} \text{ implies M1}$$

M1

$$= \frac{-6}{4} = -\frac{3}{2}$$

AG

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A1

2

$$(ii) \quad \begin{cases} y-5 \\ y+1 \end{cases} = -\frac{3}{2} \begin{cases} (x-2) \\ (x-6) \end{cases}$$

$$\text{or } y = -\frac{3}{2}x + c \text{ and attempt to find } c$$

M1

$$\Rightarrow 3x + 2y = 16$$

OE; must have integer coefficients

A1

2

- (b) (i) Gradient of perpendicular = $\frac{2}{3}$
use of $m_1 m_2 = -1$

M1

$$\Rightarrow y - 5 = \frac{2}{3}(x - 2)$$

$$3y - 2x = 11 \text{ (no misreads permitted)}$$

A1

2

- (ii) Substitute, $x = k$, $y = 7$ into their (b)(i)

or grads $\frac{7-5}{k-2} \times \frac{-3}{2} = -1$

M1

$$\Rightarrow 2 = \frac{2}{3}(k - 2) \Rightarrow k = 5$$

or Pythagoras $(k - 2)^2 = (k - 6)^2 + 8$

A1

2

[8]

Q20.

- (a) (i) Centre $(-3, 2)$
 ± 3 or ± 2

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M1

correct

A1

2

- (ii) Radius = 5

accept $\sqrt{25}$ but not $\pm \sqrt{25}$

B1

1

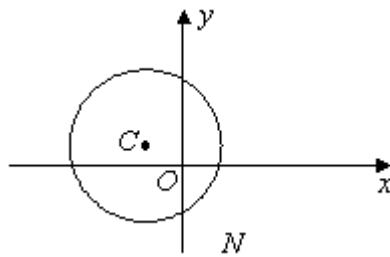
- (b) (i) $3^2 + (-4)^2 = 9 + 16 = 25$

$\Rightarrow N$ lies on circle

must have $9 + 16 = 25$ or a statement

B1

(ii)



*must draw axes;
ft their centre in correct quadrant*

M1

correct (reasonable freehand circle enclosing origin)

A1

2

(iii) Attempt at gradient of CN

withhold if subsequently finds tangent

M1

$$\text{grad } CN = -\frac{4}{3}$$

CSO

A1

$$y = -\frac{4}{3}x - 2$$

(or equivalent)
ft their grad CN

A1 ✓

3

(c) (i) $P(2, 6)$ Hence $PC^2 = 5^2 + 4^2$
"their" PC^2

M1

$$\Rightarrow PC = \sqrt{41}$$

A1

2

(ii) Use of Pythagoras correctly

M1

$$PT^2 = PC^2 - r^2 = 41 - 25,$$

where t is a point of contact of tangent

ft their PC^2 and r^2

A1 ✓

$$\Rightarrow PT = 4$$

Alternative

sketch with vertical tangent M1

showing that tangent touches circle at

point (2, 2) A1

hence $PT = 4$ A1

A1

3

[14]

Q21.

(a) $y_P = 4$

B1

1

(b) $y = 1 + \frac{2}{x} + \frac{2}{x} + \frac{4}{x^2}$

(B1 if only one error in the expansion)

For B2 the last line of the candidate's solution must be correct

B2,1,0

$$y = 1 + 4x^{-1} + 4x^{-2}$$

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2

(c) $\frac{dy}{dx} = -4x^2 - 8x^{-3}$

Index reduced by 1 after differentiating x
to a negative power

M1

At least 1 term in x correct ft on expn

A1ft

CSO Full correct solution. ACF

A1

3

(d) When $x = 2$, $\frac{dy}{dx} = -4 \times 2^{-2} - 8 \times 2^{-3}$

Attempt to find $y'(2)$.

M1

Gradient = $-1 - 1 = -2$

AG (be convinced-no errors seen)

A1

2

(e) $-2 \times m' = -1$

$m_1 \times m_2 = -1$ OE stated or used. PI

M1

$y - 4 = m(x - 2)$

*C's y_P from part (a) if not recovered;
 m must be numerical.*

M1

$y - 4 = \frac{1}{2}(x - 2)$

Ft on candidate's y_P from part (a) if not recovered.

A1ft

$x - 2y + 6 = 0$

CAO Must be this or $0 = x - 2y + 6$

A1

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4

[12]

Q22.

(a) (i) $15 + 4k = 7 \Rightarrow 4k = -8 \Rightarrow k = -2$

AG (condone verification or $y = -2$)

B1

1

(ii) $\frac{1}{2}(x_1 + x_2)$ or $\frac{1}{2}(y_1 + y_2)$

M1

Midpoint coordinates $\left(3, -\frac{1}{2}\right)$
One coordinate correct implies M1

A1

2

(b) Attempt at $\Delta y/\Delta x$ or $y = -\frac{3}{4}x + \frac{7}{4}$
(Not x over y)(may use M instead of A/B)

M1

Gradient $AB = -\frac{3}{4}$
-0.75 etc any correct equivalent

A1

2

(c) (i) $m_1 m_2 = -1$ used or stated

A1

Hence gradient $AC = \frac{4}{3}$
Follow through their gradient of AB from part (b)

A1ft

2

(ii) $y - 1 = \frac{4}{3}(x - 1)$ or $3y = 4x - 1$ etc
Follow through their gradient of AC from part (c) (i) must be normal & (1,1) used

B1ft

1

(iii) $y = 0 \Rightarrow x - 1 = -\frac{3}{4}$
Putting $y = 0$ in their AC equation and attempting to find x

M1

$$x = \frac{1}{4}$$

CSO. C has coordinates $\left(\frac{1}{4}, 0\right)$

A1

2

[10]

Q23.

(a) $\frac{dy}{dx} = \frac{3}{2}x^{\frac{1}{2}} - 3$

One term correct

M1

Both correct

A1

2

(b) (i) When $x = 0$, $\frac{dy}{dx} = -3$
Ft provided answer < 0.

B1F ✓

Eqn of tangent at O is $y = -3x$
OE Ft on $y'(0)$

B1F ✓

2

(ii) At $(9,0)$ $\frac{dy}{dx} = \frac{3}{2}(9)^{\frac{1}{2}} - 3$

Attempt to find $y'(9)$

M1

Eqn tangent at A is $y - 0 = y'(9)[x - 9]$
OE

m1

$\Rightarrow y = \frac{3}{2}(x - 9) \Rightarrow 2y = 3x - 27$

CSO. AG

A1

3

(iii) Eliminating $y \Rightarrow -6x = 3x - 27$
OE method to one variable
(eg $2y = -y - 27$)

M1

$$9x = 27 \Rightarrow x = 3$$

[A1F for each coordinate; only ft on $y = kx$ tangent in (b)(i) for $k < 0$]

A1F

$$\text{When } x = 3, y = -9\}. \quad P(3, -9)\}$$

A1F

3

(c) $\int \left(x^{\frac{3}{2}} - 3x \right) dx = \frac{2}{5} x^{\frac{5}{2}} - \frac{3x^2}{2} (+c)$
One power correct

M1

Condone absence of "+ c" and unsimplified forms

A2,1,0

3

(d) $\int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx =$
PI

B1

$$= \frac{2}{5} \times 9^{\frac{5}{2}} - \frac{3}{2} \times 9^2 - 0$$

Correct use of limits following integration

M1

$$= -24.3$$

$$\text{Area of triangle } OPA = \frac{1}{2} \times 9 \times |y_P|$$

M1

$$\text{Sh.Area} = \frac{1}{2} \times 9 \times |y_P| - \left| \int_0^9 \left(x^{\frac{3}{2}} - 3x \right) dx \right|$$

OE

M1

$$= 40.5 - 24.3 = 16.2$$

A1

5

[18]

Q24.

(a) $\frac{dy}{dt} = \frac{-2}{t^2} \quad \frac{dx}{dt} = -4$

M1A1

$$\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{1}{\frac{dx}{dt}} = \frac{1}{2t^2}$$

Use chain rule

m1

Follow on use of chain rule (if f(t))

A1F

Or eliminate t : M1 y = f(x) attempt to differentiate M1A1 chain rule

A1F reintroduce t

4

(b) $t = 2 \quad m_T = \frac{1}{8}$

follow on gradient (possibly used later)

B1F

$x = -5 \quad y = 2$

B1

$$y - 2 = \frac{1}{8} (x + 5)$$

Their (x, y), m

M1

$$x - 8y + 21 = 0$$

Ft on (x, y) and m

A1F

4

(c) $x - 3 = -4t \quad y - 1 = \frac{2}{t}$
PI

M1

$$(x - 3)(y - 1) = -4t \times \frac{2}{t} = (-8)$$

Attempt to eliminate t

M1

AG convincingly obtained

A1

3

[11]

Q25.

(a) (i) Gradient $AB = \frac{1-7}{5-1}$

Must be y on top and subtraction of coordinates

M1

$$= -\frac{6}{4} = -\frac{3}{2} = -1.5$$

Any correct equivalent

A1

2

(ii) $y - 7 = m(x - 1)$ or $y - 1 = m(x - 5)$

Verifying 2 points or $y = -\frac{3}{2}x + c$

M1

leading to $3x + 2y = 17$
AG (or grad & 1 point verified)

A1

2

(b) Attempt to eliminate x or y : $7x = 42$ etc

Solving $x - 4y = 8$; $3x + 2y = 17$

M1

$$x = 6$$

A1

$$y = -\frac{1}{2}$$

C is point $(6, -\frac{1}{2})$

A1

3

- (c) Grad of perp = $-1 / \text{their gradient } AB$
 Or $m_1 m_2 = -1$ used or stated

M1

$$= \frac{2}{3}$$

ft their gradient AB

A1 ✓

$$y - 7 = \frac{2}{3}(x - 1) \quad \text{or } 3y - 2x = 19$$

CSO Any correct form of equation

A1

3

[10]

Q26.

(a) $\frac{dy}{dx} = -10x^4$

kx^4 condone extra term

M1

Correct derivative unsimplified

A1

2

- (b) When $x = 1$, gradient = -10
 FT their gradient when $x = 1$

B1 ✓

Tangent is

Attempt at y & tangent (not normal)

M1

$$y - 5 = -10(x - 1) \text{ or } y + 10x = 15 \text{ etc}$$

CSO Any correct form

A1

(c) When $x = -2$ $\frac{dy}{dx} = -160$ (or < 0)

Value of their $\frac{dy}{dx}$ when $x = -2$

B1 ✓

($\frac{dy}{dx} < 0$ hence) y is **decreasing**

ft Increasing if their $\frac{dy}{dx} > 0$

E1 ✓

2

[7]

Q27.

- (a) (i) Attempt at $\Delta y / \Delta x$ (used with numbers)
Not x over y

M1

$$= \frac{2}{12} = \frac{1}{4}$$

0.25 etc any correct equivalent

A1

2

- (ii) $y - 2 = m(x - 11)$ or $y + 1 = m(x + 1)$

M1

$$4y - x = -3 \text{ etc leading to}$$

$$x - 4y = 3$$

or $y = mx + c$ and attempt to find c
(or sub both points into given equation)
AG (be convinced)

A1

2

- (b) Attempt to eliminate x or y

$$17y = 17 \text{ etc}$$

M1

$$y = 1$$

A1

$$x = 7$$

C is point (7,1)

A1

3

[7]

Q28.

(a) $\frac{dy}{dx} = 5x^4 - 18x^2 - 3$

Decrease one power by 1

M1

One term correct

A1

All correct

A1

3

(b) (i) Sub $x = 2$ into their $\frac{dy}{dx}$
 $80 - 72 - 3$

M1

Shown to equal 5

AG (be convinced)

A1

2

(ii) Gradient of normal $= -\frac{1}{5}(y + \frac{1}{5}x + \dots)$
Or $m_1 m_2 = -1$ used or stated

B1

$$y - 3 = -\frac{1}{5}(x - 2)$$

*Trying normal NOT tangent
 or $y = mx + c$ and attempt to find c*

M1

$x + 5y = 17$ (integer coefficients)
 Or integer multiple of coefficients

A1

3

(c) Sub $x = 1$ into their $\frac{dy}{dx}$ ($= -16 < 0$)

$$(5 - 18 - 3 = -16) \text{ (Watch } \frac{d^2y}{dx^2} = -16 \text{ !)}$$

M1

Negative value $\Rightarrow$ DECREASING

Correct interpretation of sign of $\frac{dy}{dx}$

E1ft

2

[10]

Q29.

(a) (i) $y = x + 2x^{-1}$

PI by sight of $-2x^{-2}$

B1

$$\frac{dy}{dx} = 1 - 2x^{-2}$$

One term correct

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M1

OE

A1

3

(ii) When $x = 2$, $\frac{dy}{dx} = 1 - \frac{2}{4} = \frac{1}{2}$

CSO AG (be convinced)

A1

1

(b) When $x = 2$, $y = 3$

For $y = 3$

B1

gradient of normal = - 2

$$m \times m' = -1 \text{ used}$$

M1

Equation normal $y - 3 = -2(x - 2)$

$$y - 3 = m(x - 2) \text{ OE}$$

M1

Award at 1st correct form

A1

4

[8]

Q30.

(a) Midpoint $\left(\frac{6+2}{2}, \frac{5-1}{2} \right) = (4, 2)$

One coordinate correct unsimplified

M1

Both correct and simplified

A1

2

(b) $AB^2 = (6 - 2)^2 + (5 + 1)^2$

Pythagoras used (condone one slip)

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$$= (16 + 36) = 52$$

$$52 \text{ or } \sqrt{52} \text{ seen}$$

M1

A1

$$\Rightarrow AB = 2\sqrt{13}$$

A1

3

(c) (i) Gradient $AB = (5 - -1) / (6 - 2)$

Must be y on top and subtraction (6 - 2)

M1

$$= \frac{6}{4} = \frac{3}{2} = 1.5$$

Any correct equivalent

A1

2

(ii) $y - 5 = m(x - 6)$ or $y + 1 = m(x - 2)$
or $y = mx + c$ and attempt to find c

M1

$$2y - 10 = 3x - 18 \text{ etc leading to}$$

$$3x - 2y = 8$$

AG (be convinced)

A1

2

(d) Attempt to eliminate x or y

$$7x = 28 \text{ etc}$$

$$x = 4$$

$$y = 2$$

C is point (4,2)

M1

A1

A1

3

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[12]

Q31.

(a) $(x - 2)^2 + (y + 1)^2$
 $(x \pm a)^2 + (y \pm b)^2$

M1

$$= 5^2 \text{ or } 25$$

Correct equation for circle

B1A1

3

(b) Sub $x = 6, y = 2$ into *their* circle equation
Or distance $PC^2 = 4^2 + 3^2$

M1

$$(6 - 2)^2 + (2 + 1)^2 = 16 + 9 = 25$$

Shown to equal radius²

A1

2

(c) Gradient $CP = (2 - -1) / (6 - 2)$

Must be y on top and subtraction (6 -2)

M1

$$= \frac{3}{4} = 0.75$$

Any correct equivalent

A1

2

(d) (i) Grad of perp = -1 / *their* gradient CP

Or $m_1 m_2 = -1$ used or stated

M1

$$= -4/3$$

A1

2

(ii) $y - 2 =$ their (d)(i) gradient $(x - 6)$

OE such as $3y + 4x = 30$

B1ft

1

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[10]

Q32.

(a) (i) $\sqrt{x} = x^{\frac{1}{2}}$

Accept $p = 0.5$

B1

1

(ii) $\int \sqrt{x} \, dx = \frac{x^{1.5}}{1.5} \{+c\}$

Index raised by 1

M1

Correct ft on p. Condone missing '+c'

A1ft

2

(iii) Area = $\int_1^4 \sqrt{x} \, dx$
Limits 1 and 4 PI

B1

$$\dots\dots = \frac{4^{1.5}}{1.5} - \frac{1}{1.5}$$

$$F(4) - F(1)$$

M1

$$= \frac{14}{3}$$

Accept 4.66 or better

A1

3

(b) (i) $y = x^{\frac{1}{2}} \Rightarrow \frac{dy}{dx} = \frac{1}{2} x^{-\frac{1}{2}}$
Index (p - 1) ft

M1

When $x = 4$, $y'(4) = 0.25$

Attempt to find $y'(4)$

M1

When $x = 4$, $y = 2$

B1

Equation of tangent: $y - 2 = \frac{1}{4} (x - 4)$
accept other forms

A1

4

(ii) When $x = 0$, $y = 1$ $B(0, 1)$
Subs $x = 0$ and then $y = 0$ into equation of tangent. PI

M1

When $y = 0, x = -4 \quad A(-4, 0)$

Correct ft y_B and x_A

(may be awarded as part of area calculation)

A1ft

$$\text{Area} = 0.5(1)(4) = 2$$

ft wrong sloping tangent and max of 1 further slip. Final answer must be +ve

A1ft

3

(c) Translation

'Translation'/'translate(d)'

B1

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Accept equivalent in words provided linked to 'translation/move/shift' (Note: B0B1 is possible)

B1

2

(d) $h = 1$

PI

B1

$$\text{Integral} = h/2 \{ \dots \}$$

$$\{ \dots \} = f(1) + 2[f(2) + f(3)] + f(4)$$

OE summing of areas of the three traps

M1

$$\{ \dots \} = 0 + 2(1 + \sqrt{2}) + \sqrt{3}$$

Condone 1 numerical slip

A1

$$\text{Integral} = \frac{1}{2} \{ 2(1 + 1.414\dots) + 1.732 \}$$

$$\text{Integral} = 0.5 \times 6.560\dots = 3.28 \text{ to 3sf}$$

Must be 3.28

A1CAO



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